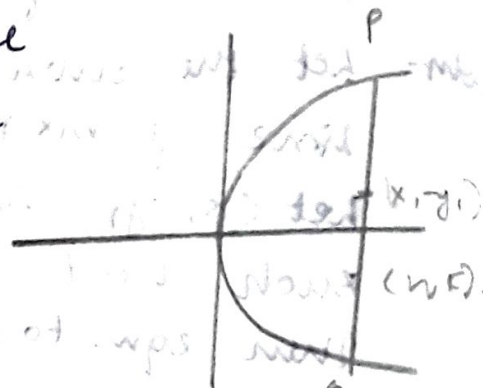


Q8

P.T.

Locus of midpts of chords of the parabola $y^2 = 4ax$ which passes through a fixed point (h, k) is the parabola $y^2 - 2ax - ky + 2ah = 0$

Soln - Let (x_1, y_1) be the midpt. of the chord PQ which passes through (h, k) . Therefore eq. to



PQ is $t_1 = S_1$

$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

$$yy_1 - 2ax = y_1^2 - 2ax_1$$

Since PQ passes through (h, k) we have

$$ky_1 - 2ah = y_1^2 - 2ax_1$$

$$y_1^2 - 2ax_1 - ky_1 + 2ah = 0$$

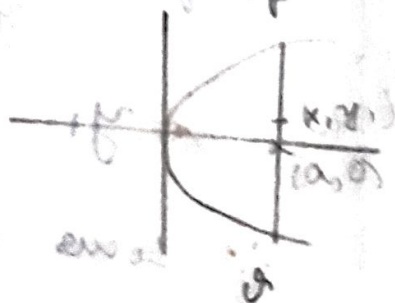
$\therefore$ Locus of (x_1, y_1) is

$$y^2 - 2ax - ky + 2ah = 0$$

Q9 Prove that the locus of the midpoints of the focal chords of the parabola $y^2 = 4ax$ is the parabola $y^2 = 2a(x-a)$.

Soln - Let (x_1, y_1) be the midpt of the chord PQ. then eq. to PQ is

$$t_1 = S_1$$



$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

$$yy_1 - 2ax = y_1^2 - 2ax_1$$

Since PQ is a focal chord, it passes through (a, 0)

$$0 - 2a^2 = y_1^2 - 2ax_1$$

~~∴~~ ∴ Locus of (x, y)

$$-2a^2 = y^2 - 2ax$$

$$y^2 = 2ax - 2a^2$$

$$y^2 = 2a(x-a)$$

Hence proved

Q11

A tangent to the parabola $y^2 + 4bx = 0$ ($y^2 = -4bx$)
meets the parabola $y^2 = 4ax$ at P and Q
show that the locus of the middle points of PQ
is $y^2(2a+b) = 4a^2x$

Soln - Given parabolas are

$$y^2 = -4bx$$

$$\text{or } y^2 = 4ax$$

let (x_1, y_1) be the midpt
of P.Q. then eq. to PQ
is

$$T_1 = S_1$$

$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

$$yy_1 - y_1^2$$

$$yy_1 - 2ax = y_1^2 - 2ax_1$$

$$yy_1 = y_1^2 + 2ax + 2a - 2ax_1$$

$$y = \frac{2ax}{y_1} + \frac{y_1^2 - 2ax_1}{y_1}$$

which is of the form $y = mx + c$

$$\text{where } m = \frac{2a}{y_1} \text{ and } c = \frac{y_1^2 - 2ax_1}{y_1}$$

Since (1) touches the parabola $y^2 = -4bx$ we have

$$c = -\frac{b}{m}$$

$$\frac{y_1^2 - 2ax_1}{y_1} = -\frac{b}{\frac{2a}{y_1}}$$

$$\frac{y_1^2 - 2ax_1}{y_1} = \frac{-by_1}{2a}$$

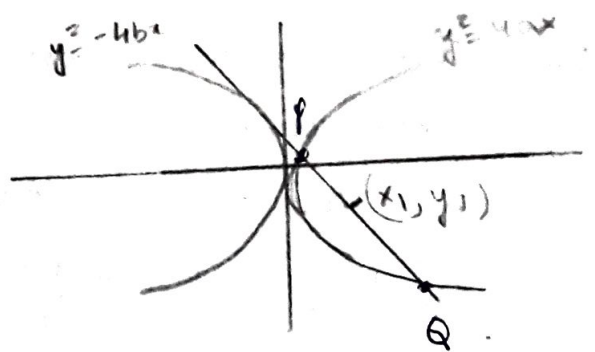
$$(y_1^2 - 2ax_1)2a = -by_1^2$$

$$2ay_1^2 - 2a^2x_1 + by_1^2 = 0$$

$$y_1^2(2a+b) = 4a^2x_1$$

∴ locus is

$$y^2(2a+b) = 4a^2x$$



Q12. Chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through a fixed point (h, k)

s.t the locus of their middle point is a curve

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{xh}{a^2} + \frac{yk}{b^2}$$

Soln- Let (x_1, y_1) be the midpoint of the chord passing through (h, k)

Eq of PQ is

$$T_1 = S_1.$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$$

Since (h, k) is a pt. on (h, k)

$$\frac{hx_1}{a^2} + \frac{ky_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$\therefore$ Locus of (x_1, y_1) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

